



the point that you need to charge your phone once every other day.

Battery types

There are several types of batteries in use today like NiMH (Nickel

Metal-hydride), NiCd (Nickel Cadmium) but the most popular would be the Li-ion (Lithium Ion) battery. It pretty much comes down to the manufacturer as NiMH and NiCd are cheap but they require frequent recharging and don't hold as much charge as LiPo (Lithium Polymer) or Li-ion. These have been proven to hold enormous amount of electric charge while reducing recharge time. In the last two years itself, companies have come up with more different types of batteries than in the past decade. However, it will be some time before we get to see these enter mainstream usage. Batteries are manufactured according to their usage conditions and a battery that works efficiently in a car may not perform at par in another form factor.

Battery rating

When you flip your mobile battery, you're bound to see a few figures here and there. These figures include certification emblems, safety standards applicable and most importantly the battery type, voltage rating and the battery capacity. A battery is called so, as it contains multiple cells, each of which is capable of carrying a certain charge of its own. However, this is very little for general use and thus multiple cells are combined in series/parallel to increase the voltage as well as current capacity.

The voltage rating simply gives the potential difference between the two terminals of the battery. This is pretty much fixed depending upon the type of chemicals used in the construction of the battery. For Li-ion batteries it can be 3.6v, 3.7v or 3.2v. Charge rating is shown using the unit mAh (milliAmpere-hour). One ampere-hour is equal to 3600 coulombs. Batteries are rated by these two parameters, i.e. voltage and current capacity (mAh).

The battery life or capacity can be calculated from the input current rating of the battery and the load current of the circuit. Battery life will be high when the load current is less and vice versa.

The battery capacity will be mentioned on the battery jacket but the load current being a variable is difficult to figure out. Since it depends on a